

Pearson Edexcel Award

Tuesday 24 June 2025

Afternoon (Time: 3 hours)

Paper
reference

9811/01

Advanced Extension Award Mathematics

Insert for questions 5, 6 and 7

Do not write on this insert.

Turn over ►

P76378A

©2025 Pearson Education Ltd.
Y:1/1/1/




Pearson

5. The discrete random variable X has the probability distribution

n	2	4	6	8	10
$P(X = n)$	$\frac{1}{10} \operatorname{cosec}^2 x$	a	$\frac{1}{5} \cot^2 x$	$\frac{1}{10} \sec^2 x$	$\frac{1}{5} \tan^2 x$

where a and x are constants.

(a) Show that

$$\sin^2 2x = \frac{6}{7 - 5a} \quad (7)$$

(b) Use the result from part (a) to find a restriction on the maximum possible value of a . (2)

The random variables X_1 and X_2 are independent and each have the same distribution as X .

Given that $P(X_1 \times X_2 = 16) = a - \frac{1}{25}$

(c) determine any possible values of a , justifying your answer. (8)
(+S2)

(Total for Question 5 is 19 marks)

6. Relative to a fixed origin O , the points A , B and C have position vectors

$$\vec{OA} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} \quad \vec{OB} = -4\mathbf{i} + 7\mathbf{j} - \mathbf{k} \quad \vec{OC} = -2\mathbf{i} + 17\mathbf{j} + 10\mathbf{k}$$

- (a) Show that triangle ABC is isosceles.

(2)

The circle S lies inside triangle ABC such that each side of the triangle is a tangent to S .

The centre, X , of S lies on the angle bisector of each vertex of the triangle.

The point D lies on AC such that $|\vec{AD}| = |\vec{AB}|$

- (b) Find the position vector of D .

(2)

- (c) Find the position vector of the midpoint of BD .

(2)

- (d) Hence show that the position of X is given by

$$\vec{OX} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} + p(-2\mathbf{i} + \mathbf{j})$$

where p is a scalar constant.

(4)

- (e) Use the angle bisector of vertex B to find another expression for $\vec{OX}$ in terms of a scalar constant q .

(3)

- (f) By equating the $\mathbf{k}$ components of the two expressions for $\vec{OX}$, determine the value of q .

(2)

- (g) Find

- (i) the position vector of the centre of S ,

(2)

- (ii) the radius of S .

(4)

(+S2)

(Total for Question 6 is 23 marks)

7. Given $u = y^x$

- (a) by taking $\ln$ of both sides, or otherwise, determine $\frac{du}{dx}$ in terms of x , y and $\frac{dy}{dx}$ only. (5)

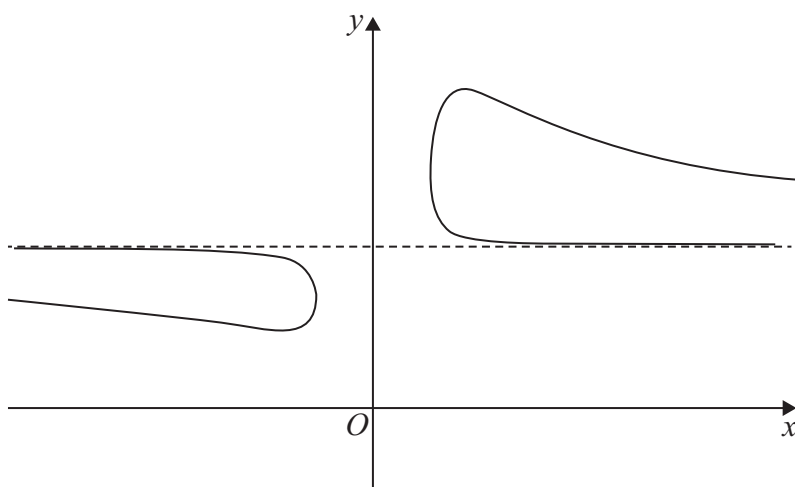


Figure 1

Figure 1 shows a sketch of the curve given by the equation

$$3y^x - x^3 \ln y = 0$$

The curve has a horizontal asymptote $y = 1$, and is defined for

- $x \leq -a, \quad x \geq a$
- $b \leq y \leq c \quad y \neq 1$

where a , b and c are positive constants.

- (b) Determine $\frac{dy}{dx}$ for this curve. (4)

- (c) Find the value of a . (6)

- (d) Find the value of b and the value of c . (7)
(+S2)

(Total for Question 7 is 24 marks)